DEEP DIVE · REFERENCE · PRINT-READY

UK Radio Frequency Allocations

Primary source: Ofcom UK Frequency Allocation Table (UKFAT) · Mode subdivisions: RSGB Band Plan 2026 · Compiled 4 Nov 2026 · Antenna lengths computed at band centre

Every frequency band allocated in the UK, what it's used for, how the spectrum is divided between modes (CW, SSB, FM, digital protocols), and the corresponding antenna lengths (half-wave and quarter-wave dipole). The **Useful Frequencies** section covers the frequencies you actually want to listen to — ADS-B, GPS, weather, aviation, DMR, APRS, FT8, PSK31, RTTY, and UK broadcast.

How to read the antenna columns: λ/2 = half-wave dipole total length (two equal elements, 180° apart, fed at centre). λ/4 = quarter-wave vertical monopole height (over ground plane) — the most common practical antenna. Values are for free space at the stated frequency; real-world shortening factor ≈ 95% for wire dipoles. For VHF+ FM work, a J-pole or 5/8-wave vertical is the usual recommendation over a bare λ/4.

01 Full UK Allocation Table — LF through UHF

Allocations verified against the Ofcom UKFAT. Mode subdivisions from the RSGB Band Plan 2026 (IARU Region 1).

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	λ/2 DIPOLE	λ/4	NOTES
148.5–255 kHz	**LF**	Longwave — BBC Radio 4 (225 kHz), time signals	`AM (DSB), 9–10 kHz ch` Groundwave only, very local — needs a large loop antenna.	681.3 m	340.7 m	λ/2 ≈ 679 m — impractical; use loop or long wire
526.5–1605 kHz	**MF — AM radio**	UK domestic AM radio (BBC Local, commercial) + BBC World Service (long-range at night)	`AM (DSB), 10 kHz ch` Day: short range · Night: skywave over hundreds of km — the classic 'whisper' band.	149.9 m	74.9 m	Long wire/vertical; λ/4 at 1 MHz = 74.95 m

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	Λ/2 DIPOLE	Λ/4	NOTES
1.6–2.0 MHz	**160m**	Amateur — long-range, night, DX	`CW (1810–1838), narrowband (1838–40), all-modes (1840–43), LSB voice (1843–2000)` • **CW** 1810–1838 kHz · **FT8/WSPR** 1838–40 · **PSK31** 1840–43 • **SSB LSB** voice 1843–2000 kHz (dial 1843 min) · QRP 1836 · DF beacons 1960	79.7 m	39.9 m	32 W max (UK); primary to foreign stations below 1850
3.5–3.8 MHz	**80m**	Amateur — workhorse DX band, night	`CW (3500–570), narrowband (3570–590), LSB voice (3600–800)` • **CW** 3500–3570 (contests 3510–555) · **FT8** 3573–75 · **PSK31** 3574 · **RTTY** 3591 (R1) • **SSB LSB** 3600–3800 (dial 3603 min) · SSB QRP 3690 · UK emergency 3663 · R1 emergency 3760	42.0 m	21.0 m	Primary; shared with other services
5.25–5.45 MHz	**60m**	Amateur — narrowband channels, Full licence only	`USB voice in 12 defined channels (all-modes per channel)` • **USB recommended** (not LSB) · CW QRP 5262 · UK emergency 5278.5 · beacon 5290 · AM 6 kHz 5317 • WSPR 5362–5374.5 · channels 5258.5 / 5276 / 5288.5 / 5298 / 5313 / 5333 / 5354 / 5362 / 5378 / 5395 / 5403.5	28.3 m	14.1 m	Full licence only, 100 W max, specific channel plan

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	Λ/2 DIPOLE	Λ/4	NOTES
7.0–7.2 MHz	**40m**	Amateur — most popular UK band, day & night DX	`CW (7000–40), narrowband (7040–50), LSB voice (7050–200)` • **CW** 7000–7040 · **FT8** 7074–76 · **PSK31** 7031 · **RTTY** 7091 (R1) • **SSB LSB** 7053–7200 (dial 7053 min) · SSB QRP 7090 · UK emergency 7110 · SSTV 7175	21.2 m	10.6 m	Primary
10.1–10.15 MHz	**30m**	Amateur — narrowband only (secondary)	`CW (10100–130) · narrowband (10130–150)` • **FT8** 10.136 · **PSK31** 10.137 • No SSB except life-safety emergencies; no contests	14.8 m	740 cm	—
14.0–14.35 MHz	**20m**	Amateur — daytime DX, 'the' band	`CW (14000–070), narrowband (14070–89), USB voice (14112–350)` • **CW** 14000–14070 · **FT8** 14074 · **PSK31** 14031 · **RTTY** 14095 (R1) · beacon 14100 • **SSB USB** 14112–14350 (contests 14125–300) · SSB QRP 14285 · SSTV 14230 · global emergency 14300	10.5 m	527 cm	Primary (satellite primary below 14250)
18.068–18.168 MHz	**17m**	Amateur — DX, less crowded	`CW (18068–095), narrowband (18095–109), USB voice (18111–168)` • **CW** 18068–18095 · **FT8** 18100 · **PSK31** 18100 · **RTTY** 18110 (R1) · beacon 18110 • **SSB USB** 18111–18168 · SSB QRP 18130 · SSTV 18165	827 cm	414 cm	Primary

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	Λ/2 DIPOLE	Λ/4	NOTES
21.0–21.45 MHz	**15m**	Amateur — great summer daytime DX	`CW (21000–070), narrowband (21070–149), USB voice (21151–450)` • **CW** 21000–21070 · **FT8** 21074–76 · **PSK31** 21316 · **RTTY** 21095 (R1) · beacon 21150 • **SSB USB** 21151–21450 · SSB QRP 21285 · SSTV 21340 · global emergency 21360 · sat 21125–245	704 cm	352 cm	Primary
24.89–24.99 MHz	**12m**	Amateur — narrowband only, no contests	`CW (24890–915), narrowband (24915–929), USB voice (24931–990)` • **FT8** 24916 · **PSK31** 24916 · **RTTY** 24920 (R1) · beacon 24930 • **SSB USB** 24931–24990 · SSB QRP 24950	601 cm	301 cm	Primary
28.0–28.5 MHz	**10m**	Amateur — summer DX, sporadic-E	`CW (28000–070), narrowband (28070–190), USB voice (28190–500), FM (28.3–4 MHz)` • **CW** 28000–28070 · **FT8** 28074–76 · **PSK31** 28125 · **RTTY** 28120 (R1) · beacon 28190 • **SSB USB** 28190–28500 · SSB QRP 28330 · SSTV 28220 · **FM voice** 28.30–28.40 MHz (50 kHz ch)	532 cm	266 cm	Primary; 30 W for Intermediate

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	Λ/2 DIPOLE	Λ/4	NOTES
50–52 MHz	**6m**	Amateur — tropospheric ducting DX in summer	**CW** (50.0–01) · **SSB USB** (50.1–04) · FM (50.7+) • **CW** 50.000–50.100 (beacons 50.000–030) · **SSB** 50.100–50.300 (intl CoA 50.150) · MGM 50.305 • Beacons 50.400–50.500 (no TX) · **FM voice** 51.410–51.590 (calling **51.510**) · repeaters 51.220–51.370 in / 50.720–50.870 out	298 cm	149 cm	Primary · FM: J-pole ≈ 4.1 m or 5/8-wave ≈ 3.7 m (preferred over λ/4)
70–70.5 MHz	**4m**	Amateur — NoV band (Full licence)	Narrowband (70.1–25) · 12.5 kHz ch all-modes (70.294–500) • Beacons 70.000–70.090 · **CW/SSB centre** 70.200 · **AM/FM calling** 70.260 · MGM 70.270 • **FM** 70.294–70.500 (2.5 kHz steps): internet gateways 70.350/362.5/387.5/412.5 · **FM calling** 70.450 · GB2RS 70.425	213 cm	107 cm	Secondary; NoV required, 160 W max · FM: J-pole ≈ 2.9 m or 5/8 ≈ 2.7 m (preferred over λ/4)

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	Λ/2 DIPOLE	Λ/4	NOTES
144–146 MHz	**2m**	Amateur VHF — FM voice, SSB, digital	**FM voice** (145.0–07) · **SSB** (144.15–40) · CW (144.0–15) • **CW** 144.025–144.150 (CoA 144.050) · **SSB USB** 144.150–144.400 (CoA 144.300, GB2RS 144.250) • **FM** simplex 145.200–145.5935 (calling **145.500**) · **APRS** 145.525 (AX.25 1200 bd AFSK) • **DMR** 145.275 · **C4FM** 145.2375 · repeater inputs 145.100–145.500 / outputs 145.500–146.000 (8 MHz split)	103 cm	51.5 cm	Primary; 400 W ERP · FM: J-pole ≈ 1.4 m or 5/8 ≈ 1.3 m (preferred over λ/4); 1/2-wave collinear for mobile
430–440 MHz	**70cm**	Amateur UHF — FM, DV, digital, EME	**FM/DV** (430.1–431.0) · **SSB/CW** (431.0–432.4) · FM (433.0+) • **CW/SSB** 431.000–432.400 (2.7 kHz) · **FM voice** 433.000–433.500 (calling **433.500**) · **DMR** 433.500 • Repeater outputs 438.000–439.000 / inputs 430.000–431.000 (7.6 MHz split) · sat uplinks 435–438 · EME CW 432.0–432.1	34.3 cm	17.1 cm	Primary; 400 W ERP; conditions in 430–432 MHz · FM: J-pole ≈ 47 cm or 5/8 ≈ 43 cm (preferred over λ/4)

Sideband rule of thumb: below 10 MHz use **LSB**, above 10 MHz use **USB**. Minimum dial settings: 1843 kHz (160 m), 3603 kHz (80 m), 7053 kHz (40 m).

02 Useful Frequencies — Listen / Receive

The frequencies worth tuning to, with mode and antenna. Sorted by service, not frequency.

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	λ/2 DIPOLE	λ/4	NOTES
1090 MHz	**ADS-B (Mode S)**	Aircraft position/altitude/speed — Mode S 1090ES, one receiver does everything	`Pulse 1 Mbaud, 8-bit`	13.8 cm	6.88 cm	Small V-band patch or 5/8 whip; λ/2 = 13.6 cm
156.050 MHz	**Ch 16 — DSC**	Marine VHF distress & calling (digital)	`DSC digital + GMDSS voice`	96.1 cm	48.0 cm	Whip λ/4 ≈ 47.6 cm (mobile marine mast)
156.525 MHz	**Ch 85B — DSC**	Marine DSC digital distress/calling	`DSC digital`	95.8 cm	47.9 cm	—
121.5 MHz	**Aviation emergency**	Guard frequency — aircraft emergency (ELT)	`AM voice, 25 kHz ch`	123 cm	61.7 cm	—
122.900 MHz	**Air Traffic (UK)**	UK ATC control / en-route (example)	`AM voice, 25 kHz ch`	122 cm	61.0 cm	Aviation band 118–137 MHz — AM, not FM
108–117.975 MHz	**VOR / ILS glideslope**	VOR navigation + ILS glideslope (UK)	`AM / FM 30 Hz/90 Hz`	134 cm	66.9 cm	Sits inside the FM broadcast band — VOR/ILS on centre kHz
108–118 MHz	**ILS localiser**	Instrument landing — localiser (within VOR band)	`AM, 100 Hz mod`	136 cm	68.1 cm	—
243.0 MHz	**Aviation emergency**	Guard frequency — aircraft emergency (ELT, 2m satellite link)	`AM voice, 25 kHz ch`	61.7 cm	30.8 cm	—
1027–1150 MHz	**DME**	Distance Measuring Equipment — paired with VOR/ILS (channels 1–126)	`UHF pulse pairs, 63.5 µs`	13.8 cm	6.88 cm	λ/2 ≈ 13.6 cm (1 GHz) — whip or patch

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	λ/2 DIPOLE	λ/4	NOTES
1575.42 MHz	**GPS L1 C/A**	Civil GPS positioning	`CDMA, 1.023 Mcps`	9.51 cm	4.76 cm	Patch antenna (V-band) — λ/2 ≈ 9.5 cm
1227.60 MHz	**GPS L2**	GPS L2 (military L2C + civil)	`CDMA`	12.2 cm	6.11 cm	—
1176.45 MHz	**GPS L5**	GPS L5 — improved accuracy	`CDMA`	12.7 cm	6.37 cm	—
1191.795 MHz	**Galileo E1**	EU GNSS (shared L-band)	`CDMA`	12.6 cm	6.29 cm	—
1207.14 MHz	**Galileo E5a**	EU GNSS — safety-of-life	`CDMA`	12.4 cm	6.21 cm	—
1544.43 MHz	**Galileo E5b/E6**	EU GNSS E5b (PPP) / E6	`CDMA`	9.71 cm	4.85 cm	—
1598–1605.89 MHz	**GPS L1C (new)**	Next-gen civilian L1C	`CDMA`	9.36 cm	4.68 cm	—
1602 MHz	**GLONASS G1**	Russian GNSS (L1, 1602–1605.4 MHz)	`CDMA/FDMA`	9.35 cm	4.68 cm	—
403–405 MHz	**Weather — MAS band**	Meteorological Aids Service band (weather datacasting, aeronautical METAR feeds)	`Digital subcarrier / FSK`	37.1 cm	18.6 cm	λ/4 ≈ 18.6 cm whip
162.400–162.550 MHz	**Weather — US NOAA**	US NWS weather radio (WXR) — 7 channels	`AM + subcarrier FSK (1785 Hz)`	92.3 cm	46.1 cm	—
145.275 MHz	**DMR (2m)**	Digital Mobile Radio voice — BrandMeister / DMRplus	`DMR Tier 2, 4.5 kHz`	103 cm	51.6 cm	—
433.500 MHz	**DMR (70cm)**	DMR voice — UK hotspots	`DMR Tier 2`	34.6 cm	17.3 cm	—
145.2375 MHz	**C4FM (2m)**	C4FM digital voice (Yaesu Fusion, Icom IDAS)	`C4FM 6.25 kHz`	103 cm	51.6 cm	—

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	λ/2 DIPOLE	λ/4	NOTES
145.525 MHz	**APRS**	Amateur Packet Radio — position/status (AX.25 AFSK 1200 bd)	AX.25 / APRS	103 cm	51.5 cm	—
14.074 MHz	**FT8 (20m)**	Most popular HF digimode — weak-signal digital	FT8 (JT9-based, 6 FSK)	10.7 m	533 cm	—
7.074 MHz	**FT8 (40m)**	FT8 — 40 m	FT8	21.2 m	10.6 m	—
7.031 MHz	**PSK31 (40m)**	Slow PSK31 data	PSK31, 31 Bd	21.3 m	10.7 m	—
14.130 MHz	**PSK31 (20m)**	PSK31 — 20 m	PSK31	10.6 m	530 cm	—
7.091 MHz	**RTTY (40m)**	RTTY (R1: TX 7.091 MHz, shift 170 Hz)	FSK 170 Hz, 45 Bd	21.1 m	10.6 m	—
14.095 MHz	**RTTY (20m)**	RTTY — 20 m	FSK 170 Hz	10.6 m	532 cm	—
14.100 MHz	**Beacon (20m)**	IARU propagation beacon (no TX!)	CW 75 wpm	10.6 m	532 cm	—
28.190 MHz	**Beacon (10m)**	Propagation beacon (no TX!)	CW	532 cm	266 cm	—
145.500 MHz	**FM calling (2m)**	VHF FM voice calling / simple channel	FM 12.5 kHz	103 cm	51.5 cm	—
433.500 MHz	**FM calling (70cm)**	UHF FM voice calling	FM 12.5 kHz	34.6 cm	17.3 cm	—
51.510 MHz	**FM calling (6m)**	6 m FM voice calling	FM 12.5 kHz	291 cm	146 cm	—
70.260 MHz	**AM/FM calling (4m)**	4 m AM/FM calling	AM/FM	213 cm	107 cm	—
526.5–1605 kHz	**AM radio (UK)**	BBC Local + commercial AM	AM (DSB) 10 kHz	149.9 m	74.9 m	—

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	Λ/2 DIPOLE	Λ/4	NOTES
87.5–108 MHz	**FM radio (UK)**	UK FM broadcasting (stereo)	`FM stereo 19 kHz`	150 cm	74.9 cm	λ/2 ≈ 1.49 m, λ/4 ≈ 0.75 m (whip)
174–240 MHz	**DAB / DAB+ (UK)**	UK digital radio — Band III	`COFDM, 1.536 Mcs`	74.6 cm	37.3 cm	UK: ch 5A BBC, 6A BBC, 7A BBC, 8A/10A/11A commercial — typical 201 MHz
470–694 MHz	**UHF TV (UK)**	UK Freeview — ch 21–60 (DVB-T2)	`DVB-T2 8k, ~7 MHz`	27.3 cm	13.6 cm	Grouped UHF log-periodic or Yagi; λ/4 at 550 MHz ≈ 13.6 cm
470–790 MHz	**4G / 5G (UK)**	UK mobile: 4G 800/1800/2600 MHz, 5G 3.4–3.8 GHz + 26 GHz	`OFDMA (LTE/5G NR)`	18.7 cm	9.37 cm	—
880–915 MHz	**GSM 900 (UK)**	UK 2G/3G/4G — 900 MHz block	`GSM / WCDMA / LTE`	16.8 cm	8.42 cm	—
1805–1880 MHz	**1800 MHz (UK)**	UK 2G/3G/4G — 1.8 GHz block	`GSM / WCDMA / LTE`	8.15 cm	4.07 cm	—
2100–2200 MHz	**2100 MHz (UK)**	UK 3G/4G — 2.1 GHz block	`WCDMA / LTE`	6.97 cm	3.49 cm	—
2655–2690 MHz	**2600 MHz (UK)**	UK 4G/5G — 2.6 GHz block	`LTE / 5G NR`	5.61 cm	2.81 cm	—
3400–3800 MHz	**5G (UK)**	UK 5G mid-band (C-band)	`5G NR`	4.28 cm	2.14 cm	λ/2 ≈ 4.28 cm — patch/panel antenna

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	Λ/2 DIPOLE	Λ/4	NOTES
11.7–12.2 GHz	**Ku-band SAT TV**	Sat TV (SES-2, etc.) — downlink 10.7–12.75 GHz	QPSK/8PSK DVB-S2	1.26 cm	0.63 cm	Offset-fed dish (60–90 cm); feed LNB ≈ 2.5 cm waveguide
4.7–6.9 GHz	**C-band**	Satcom / backhaul (uplink 5.925–6.425)	QPSK	2.50 cm	1.25 cm	—

03 Known Frequencies Worth Tuning

Real, well-known frequencies across the bands for particular purposes — weather, number stations, time signals, emergency/distress, and international broadcasting. These are for **listening** (SDR or scanner) unless noted as ham emergency CoAs. Emergency frequencies are for receiving/monitoring; never transmit on them outside a genuine emergency.

Weather stations (NWS HF, worldwide SDR listening)

US National Weather Service HF weather — AM voice + digital subcarrier, receivable worldwide. The classic "weather station" band. UK has no equivalent HF service; the UK Met Office weather is on FM/TV/web.

FREQUENCY	NAME / CALLSIGN	MODE	WHAT IT IS
4125 kHz	**4125 kHz — NWS HF WX**	AM + subcarrier	NWS HF weather (also the 4 MHz HF maritime distress voice freq)
5850 kHz	**5850 kHz — NWS HF WX**	AM + subcarrier	NWS HF weather, 6 m
6315 kHz	**6315 kHz — NWS HF WX**	AM + subcarrier	NWS HF weather, 6 m
8425 kHz	**8425 kHz — NWS HF WX**	AM + subcarrier	NWS HF weather, 8 m
9200 kHz	**9200 kHz — NWS HF WX**	AM + subcarrier	NWS HF weather, 9 m
12630 kHz	**12630 kHz — NWS HF WX**	AM + subcarrier	NWS HF weather, 13 m
14640 kHz	**14640 kHz — NWS HF WX**	AM + subcarrier	NWS HF weather, 20 m

Number stations (spy / mystery shortwave)

One-way coded transmissions, usually synthetic voice reading groups of five digits, often introduced by a distinctive tune. No published schedules; frequencies and times shift without notice. From the 2026 B-26 EIBI listener-log schedule — treat as a guide, not gospel.

FREQUENCY	NAME / CALLSIGN	MODE	WHAT IT IS
3168 kHz	**Channel Marker L**	CW (Morse)	Continuous, 24 h — Russia
3243 kHz	**Goose Channel Marker**	Voice / various	Nightly UTC, Russia
3364 kHz	**The Squeaky Wheel**	Voice / various	Nightly UTC, Russia
3850 kHz	**Baron Channel Marker**	Voice / various	Nightly UTC, Russia
4770 kHz	**Alarm Channel Marker**	Voice / various	Continuous, Russia
4909 kHz	**E11 (Polish spy numbers)**	Synthetic voice	Multiple UTC slots, Poland
5082 kHz	**E11 (Polish spy numbers)**	Synthetic voice	Evening UTC, Poland
5943 kHz	**S06 (Russian spy numbers)**	Synthetic voice	Evening UTC, Russia
6804 kHz	**E11 (Polish spy numbers)**	Synthetic voice	Morning UTC, Poland
6849 kHz	**E11 (Polish spy numbers)**	Synthetic voice	Evening UTC, Poland

Time & frequency standard signals

National time/frequency beacons — great for checking a receiver against the standard. WWVB/WWV are NIST (US).

FREQUENCY	NAME / CALLSIGN	MODE	WHAT IT IS
60 kHz	**WWVB (NIST)**	CW + digital code	LF, US radio clocks — needs a loop antenna
100 kHz	**WWV (NIST)**	CW + digital code	LF time/frequency standard
2.5 MHz	**WWV / WWVH**	CW + digital code	HF time code, NIST
5 MHz	**WWV / WWVH**	CW + digital code	HF time code, NIST
10 MHz	**WWV / WWVH**	CW + digital code	HF time code, NIST
15 MHz	**WWV / WWVH**	CW + digital code	HF time code, NIST
20 MHz	**WWV**	CW + digital code	HF time code (WWV only)

Emergency, distress & safety (GMDSS + aviation)

International distress and safety frequencies — these are the ones a scanner/SDR should always keep programmed. Do NOT transmit on these unless in an actual emergency.

FREQUENCY	NAME / CALLSIGN	MODE	WHAT IT IS
2182 kHz	**MF distress (radiotelephony)**	`AM voice`	International MF distress; hourly 3-min silence periods
4125 kHz	**4 MHz HF distress voice**	`AM voice`	HF maritime distress (also NWS weather)
4207.5 kHz	**4 MHz HF DSC**	`Digital`	Maritime digital selective calling
518 kHz	**NAVTEX**	`Digital FSK`	International maritime safety information (primary)
490 kHz	**NAVTEX (regional)**	`Digital FSK`	Regional NAVTEX
6215 kHz	**6 MHz HF distress voice**	`AM voice`	HF maritime distress
8291 kHz	**8 MHz HF distress voice**	`AM voice`	HF maritime distress
12290 kHz	**12 MHz HF distress voice**	`AM voice`	HF maritime distress
16420 kHz	**16 MHz HF distress voice**	`AM voice`	HF maritime distress
121.5 MHz	**Aviation emergency / EPIRB homing**	`AM`	Guard freq — aircraft distress; EPIRB locator beacon
243.0 MHz	**Aviation emergency (satellite)**	`AM`	Aircraft emergency / ELT via 2 m satellite
406.0–406.1 MHz	**EPIRB (Cospas-Sarsat)**	`Digital`	Satellite emergency beacon — position relay
156.800 MHz	**Ch 16 — marine VHF distress**	`AM voice / DSC`	Continuous maritime listening watch
156.525 MHz	**Ch 70 — marine DSC**	`Digital`	Maritime digital distress, no voice
9200–9500 MHz	**SART (X-band)**	`Radar transponder`	Search & Rescue transponder — 12 dots on radar

UK / R1 amateur emergency centres (for ham operators)

Designated emergency CoAs from the RSGB band plan — where hams go to handle emergency traffic. These sit inside the ham allocations already listed in section 1.

FREQUENCY	NAME / CALLSIGN	MODE	WHAT IT IS
5278.5 kHz	**60 m UK emergency**	`SSB / CW`	UK emergency comms traffic, 60 m
3663 kHz	**80 m UK emergency**	`SSB LSB`	UK emergency comms, 80 m
7110 kHz	**40 m R1 emergency**	`SSB LSB`	IARU Region 1 emergency CoA, 40 m

FREQUENCY	NAME / CALLSIGN	MODE	WHAT IT IS
14300 kHz	**20 m global emergency**	SSB USB	Global emergency CoA, 20 m
21360 kHz	**15 m global emergency**	SSB USB	Global emergency CoA, 15 m
145.500 MHz	**2 m VHF emergency**	FM / DMR	VHF FM calling / emergency, 2 m
433.500 MHz	**70 cm VHF emergency**	FM / DMR	UHF FM calling / emergency, 70 cm

International shortwave broadcasting

Well-known international broadcasters (BBC, VOA, etc.) — these change seasonally with propagation, so treat exact kHz as a starting point. For current schedules use the BBC Global Shortwave Frequency page or Shortwave DB. The bands with the most international voice are 3.5, 7, 11, 15 and 17 MHz.

FREQUENCY	NAME / CALLSIGN	MODE	WHAT IT IS
3.5–4 MHz	**International broadcasting (AM)**	AM voice	Shortwave — many international stations
7–9 MHz	**International broadcasting (AM)**	AM voice	Shortwave — heavy international use
11–12 MHz	**International broadcasting (AM)**	AM voice	Shortwave — daytime international
15–16 MHz	**International broadcasting (AM)**	AM voice	Shortwave — daytime international
17–18 MHz	**International broadcasting (AM)**	AM voice	Shortwave — daytime international

04 Mode & Protocol Glossary

MODE	DESCRIPTION
CW	Continuous wave — Morse code (telegraphy). 200 Hz bandwidth, narrowest & most sensitive mode.
LSB	Lower Sideband SSB voice — used below 10 MHz (160/80/40 m). Dial setting = carrier, subtract ~3 kHz for tone.
USB	Upper Sideband SSB voice — used above 10 MHz (30/20/17/15/12/10 m).
AM / DSB	Amplitude Modulation, double sideband — conventional AM radio, aviation voice.
FM	Frequency Modulation — 2m/70cm/6m simplex & repeaters, commercial FM radio, 12.5 kHz channels.
J-pole	End-fed vertical: ¼λ radiator + ⅜λ folded feed stub, fed at the bottom = 11/16λ total. No ground plane needed (self-resonant), ~1 dB over a λ/4, easy to build from a single wire. The go-to portable/mobile FM vertical.

MODE	DESCRIPTION
5/8-wave	5/8λ vertical over a ground plane (radials). ~2–3 dB gain over λ/4, slightly lower feedpoint impedance — the standard for fixed VHF/UHF base stations and repeaters.
FT8	Weak-signal digital (WSJT-X) — 6-tone FSK, ~6.7 Hz bandwidth, 15 s cycles. The default HF digimode.
FT4	Faster version of FT8 — 7.5 s cycles, 4 FSK tones.
WSPR	Very weak-signal propagation (WSJT-X) — 4 FSK tones, 2.5 Hz bandwidth, 2-min cycles. Listen in the CW segment; TX anywhere in the narrowband sub-band.
PSK31	Phase-shift keying, 31 baud — keyboard data, ~31 Hz bandwidth. Popular on 40/20 m.
RTTY	Ripple Telegraphy — FSK 170 Hz shift, 45 baud. TX frequency given is the mark (audio 1700 Hz).
AMTOR / Hellschreiber	Older robust digimodes — still on the band plan, rarely used now.
DMR	Digital Mobile Radio — 2m/70cm digital voice, 4.5 kHz, BrandMeister/DMRplus repeater network.
C4FM	C4FM digital voice (Yaesu Fusion / Icom IDAS) — 6.25 kHz, 2 m.
AX.25 / APRS	Packet radio / Automatic Position Reporting — 1200 baud AFSK on 145.525 MHz (UK).
MGM	Machine-Generated Mode — umbrella for computer-driven digimodes.
CoA	Centre of Activity — recommended calling frequency for a mode/segment.
SSTV	Slow-Scan TV — image transmission, 40 m (7.175 MHz), 20 m (14.230 MHz).
DSC	Digital Selective Calling — marine digital distress/calling (Ch 16 / 85B).
ADS-B	Automatic Dependent Surveillance-Broadcast — aircraft position via Mode S 1090 MHz.

05 Sources

- Ofcom — UK Frequency Allocation Table (UKFAT): ofcom.org.uk (PDF, 253 pp)
- RSGB — Band Plan 2026 (effective 1 Jan 2026): rsgb.org
- Ofcom — Spectrum Information System (SIS): ofcom.org.uk
- Antenna formulas: λ/2 = c / 2f, λ/4 = c / 4f, c = 299,792,458 m/s